

Reproductive and Sexual Health

Fertility, menstrual and gonadal physiology, sexual function, contraception, pelvic health, measurement, and rights

Volume 03M of the Health Atlas Research Suite

Purpose

Integrate reproductive biology with sexual function, autonomy, relationships, public health, devices and longitudinal life-course care.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume is not individualized fertility, contraception, hormone or STI advice. Pregnancy emergencies, severe bleeding, testicular torsion, acute pelvic infection and sexual violence require immediate qualified pathways. Intimate data and examinations require explicit consent and privacy.

How to use this volume

Reproductive and sexual health encompasses bodily function, fertility and infertility, contraception, sexual well-being, prevention and treatment of infection, pelvic health, autonomy, safety and the ability to pursue or avoid pregnancy. It is not reducible to reproductive organs or hormones: vascular, neural, endocrine, immune, relational, cultural, legal and economic conditions shape outcomes and access.

Questions this volume is designed to answer

1. How do reproductive, sexual, endocrine, vascular, neural and relational systems interact?
2. What can reserve, semen and cycle tests actually predict?
3. How should fertility and sexual-health devices be validated?
4. How can care preserve autonomy and avoid coercion or stigma?
5. Which reproductive outcomes require cross-reference to maternal, developmental and public-health volumes?

Scope

- HPG regulation, gametogenesis, cycles, fertilization, sexual response and pelvic floor.
- Infertility, PCOS, endometriosis, bleeding, STI, sexual dysfunction, menopause and cancers.
- Hormonal, semen, imaging, microbiologic, genetic, functional and digital measurement.
- Contraception, fertility treatment, rehabilitation, surgery, infection/cancer prevention and rights-based care.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

Contents

Document status and intended use	i
How to use this volume	ii
1 Executive synthesis	1
1.1 What a complete profile requires	1
2 Historical development and changing models	3
2.0.1 Fertility, contraception, childbirth, sexual behavior and reproductive anatomy were described within cultural, legal and religious systems.	4
2.0.2 Anatomy, microscopy, demographic registration and germ theory clarified organs, gametes, cycles and infection.	4
2.0.3 Hormone assays, antibiotics, contraception, ultrasound, laparoscopy and microsurgery made fertility and reproductive disease tractable.	4
2.0.4 Rights-based sexual/reproductive health, randomized evidence, STI programs and maternal-safety movements expanded the outcome beyond fertility.	4
2.0.5 ART, genomics, fertility preservation, telehealth and cycle/wearable data expanded options and measurement.	4
2.0.6 Reproductive and sexual health is a life-course system linking biology, function, goals, relationships, rights, environment and healthcare.	4
3 Normal foundations: structure, physiology, and regulation	6
3.0.1 Hypothalamic-pituitary-gonadal regulation	6
3.0.2 Oogenesis, follicular development and ovulation	6
3.0.3 Spermatogenesis and male reproductive function	7
3.0.4 Menstrual cycling and endometrial function	7
3.0.5 Fertilization, implantation and early embryonic development	7
3.0.6 Sexual response and genital function	8
3.0.7 Pelvic floor and lower reproductive tract mechanics	8
3.0.8 Contraceptive physiology and reproductive autonomy	8
3.0.9 Menopause and reproductive aging	9
3.0.10 Reproductive microbiome, infection and immune tolerance	9
3.1 Useful quantitative relationships	9
4 Determinants, heterogeneity, and causal architecture	11
4.0.1 Genetic architecture and family history	13
4.0.2 Development and life course	13
4.0.3 Nutrition and metabolism	13
4.0.4 Physical activity, conditioning, and sedentary behavior	14

4.0.5	Sleep, circadian timing, and recovery	14
4.0.6	Stress, emotion, and behavior	14
4.0.7	Social, economic, cultural, and care conditions	15
4.0.8	Environmental and occupational exposures	15
4.0.9	Health care, medication, and iatrogenic effects	15
5	Major disorders, failure modes, and clinically important phenotypes	16
5.0.1	Infertility and subfertility	16
5.0.2	Polycystic ovary syndrome	16
5.0.3	Endometriosis and adenomyosis	17
5.0.4	Uterine fibroids and structural bleeding	17
5.0.5	Abnormal uterine bleeding and amenorrhea	18
5.0.6	Sexually transmitted infections and reproductive-tract infection	18
5.0.7	Pelvic inflammatory disease and tubal disease	19
5.0.8	Erectile dysfunction	19
5.0.9	Sexual pain and penetration disorders	20
5.0.10	Pelvic organ prolapse and continence-related pelvic dysfunction	20
5.0.11	Premature ovarian insufficiency and hypogonadism	21
5.0.12	Menopause-related symptoms and genitourinary syndrome	21
5.0.13	Reproductive cancers	22
5.0.14	Gender-affirming reproductive and sexual health needs	22
6	Assessment, clinical reasoning, and longitudinal profiling	24
6.0.1	1. Question and urgency	25
6.0.2	2. History and lived phenotype	25
6.0.3	3. Examination and basic measurements	25
6.0.4	4. Targeted laboratory, imaging, and physiology	26
6.0.5	5. Function and challenge testing	26
6.0.6	6. Longitudinal integration	26
7	Measurement and device landscape	27
7.0.1	Reproductive and sexual history	29
7.0.2	Hormone assays	29
7.0.3	Semen analysis	30
7.0.4	Pelvic and transvaginal ultrasound	30
7.0.5	Tubal and uterine cavity testing	31
7.0.6	Ovarian reserve testing	31
7.0.7	Ovulation and fertility tracking	31
7.0.8	Genetic and carrier testing	32
7.0.9	Sexual-function and pelvic-floor measures	32

7.0.10 Nocturnal penile tumescence and vascular testing	32
7.0.11 STI and genital microbiology testing	33
7.0.12 Endoscopy, laparoscopy and biopsy	33
7.0.13 Reproductive digital records and decision support	34
8 Engineering architectures, signal quality, and validation	35
8.0.1 Intended use and measurand	36
8.0.2 Sensor and transduction	36
8.0.3 Timing and synchronization	36
8.0.4 Analog, acquisition, and digital integrity	37
8.0.5 Calibration and traceability	37
8.0.6 Artifact and quality channels	37
8.0.7 Algorithms and uncertainty	38
8.0.8 Safety, usability, cybersecurity, and privacy	38
9 Prevention, treatment, rehabilitation, and system intervention	39
9.0.1 Comprehensive sexuality education and preventive services	40
9.0.2 Contraception	41
9.0.3 Fertility optimization and ovulation induction	41
9.0.4 Intrauterine insemination and assisted reproductive technology	41
9.0.5 Fertility preservation	42
9.0.6 Hormonal and nonhormonal reproductive therapy	42
9.0.7 Pelvic-floor rehabilitation and sexual therapy	42
9.0.8 Surgery and minimally invasive procedures	42
9.0.9 STI vaccination, treatment and partner services	43
9.0.10 Cancer prevention, treatment and sexual/fertility survivorship	43
9.0.11 Rights-based and trauma-informed care	43
10 What is known, not known, and frequently overstated	44
10.0.1 Can ovarian aging be predicted precisely for an individual?	45
10.0.2 Do reproductive microbiome tests improve outcomes?	45
10.0.3 How much risk is genetic versus environmental?	46
10.0.4 Can one composite score summarize health?	46
10.0.5 When is earlier detection beneficial?	46
10.0.6 Do digital biomarkers generalize?	47
10.0.7 What counts as meaningful change?	47
11 Research and development roadmap	48
11.0.1 0. Couple-level fertility model	49
11.0.2 0. Privacy-preserving reproductive sensor standard	49
11.0.3 1. Reference map	49

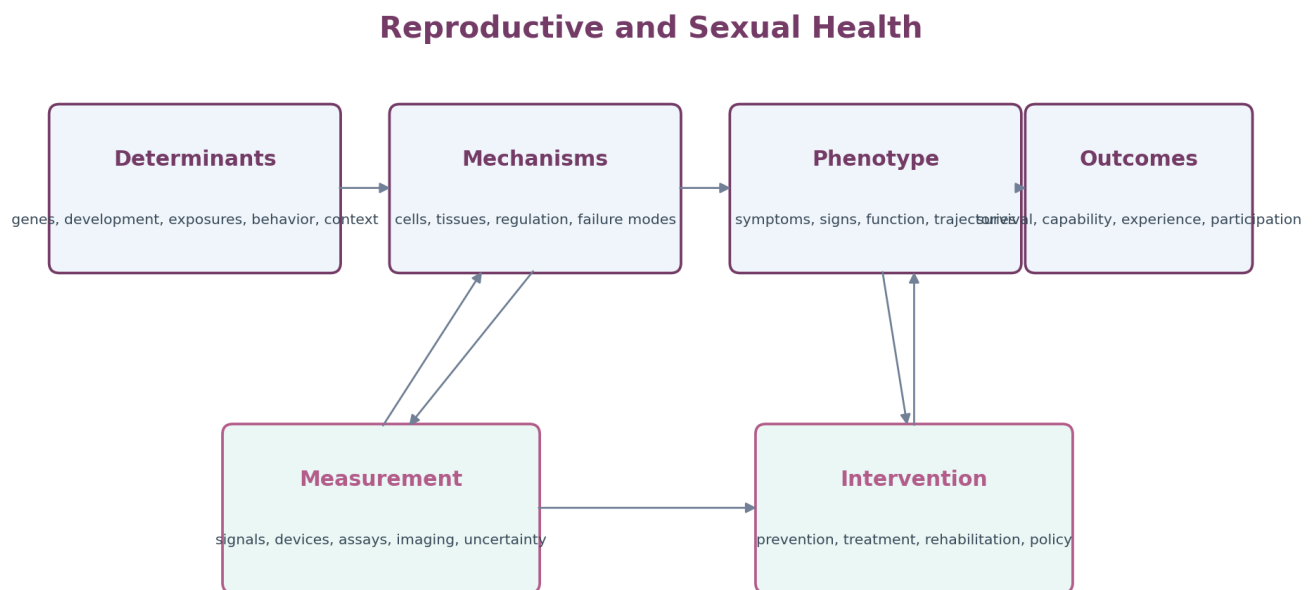
11.0.4 2. Causal determinants	50
11.0.5 3. Measurement validity	50
11.0.6 4. Clinical utility	50
11.0.7 5. Longitudinal multimorbidity	51
11.0.8 6. Implementation and governance	51
12 Cross-volume integration	52
A Minimum data schema for a longitudinal domain profile	53
B Glossary	54
Selected references and current standards	55

Executive synthesis

Reproductive and sexual health encompasses bodily function, fertility and infertility, contraception, sexual well-being, prevention and treatment of infection, pelvic health, autonomy, safety and the ability to pursue or avoid pregnancy. It is not reducible to reproductive organs or hormones: vascular, neural, endocrine, immune, relational, cultural, legal and economic conditions shape outcomes and access.

Central organizing principle

Integrate reproductive biology with sexual function, autonomy, relationships, public health, devices and longitudinal life-course care.



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Acute severe pelvic/testicular pain, heavy bleeding with instability, fever with pelvic pain, suspected ectopic pregnancy, priapism, sexual assault or airway/systemic allergic reaction requires urgent care. Intimate examinations and device data require explicit consent, confidentiality and trauma-informed practice.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early knowledge and practice	Fertility, contraception, childbirth, sexual behavior and reproductive anatomy were described within cultural, legal and religious systems.	Preserved practical knowledge and recognized reproduction as socially embedded.	Coercion, gender hierarchy and weak biological understanding often dominated.
Anatomical and demographic era	Anatomy, microscopy, demographic registration and germ theory clarified organs, gametes, cycles and infection.	Enabled causal localization and population measurement.	Women and marginalized groups were often studied without autonomy or representation.
Hormonal, laboratory and surgical era	Hormone assays, antibiotics, contraception, ultrasound, laparoscopy and microsurgery made fertility and reproductive disease tractable.	Created effective prevention, diagnosis and treatment.	Specialty silos separated sexual well-being, mental health and social determinants.
Rights, public-health and evidence era	Rights-based sexual/reproductive health, randomized evidence, STI programs and maternal-safety movements expanded the outcome beyond fertility.	Centered autonomy, prevention and population equity.	Legal and financing systems still restrict access and choice.
Molecular, assisted and digital era	ART, genomics, fertility preservation, telehealth and cycle/wearable data expanded options and measurement.	Enabled individualized pathways and longitudinal data.	Commercial claims, intimate-data governance and unequal access create new risks.
Current systems model	Reproductive and sexual health is a life-course system linking biology, function, goals, relationships, rights, environment and healthcare.	Supports whole-person and cross-volume decisions.	Classification and research still undermeasure pleasure, coercion, disability and diverse identities.

2.0.1 Fertility, contraception, childbirth, sexual behavior and reproductive anatomy were described within cultural, legal and religious systems.

Scientific consequence. Preserved practical knowledge and recognized reproduction as socially embedded.

What remained unresolved. Coercion, gender hierarchy and weak biological understanding often dominated.

Evidence anchors: [3,6]

2.0.2 Anatomy, microscopy, demographic registration and germ theory clarified organs, gametes, cycles and infection.

Scientific consequence. Enabled causal localization and population measurement.

What remained unresolved. Women and marginalized groups were often studied without autonomy or representation.

Evidence anchors: [3,6]

2.0.3 Hormone assays, antibiotics, contraception, ultrasound, laparoscopy and microsurgery made fertility and reproductive disease tractable.

Scientific consequence. Created effective prevention, diagnosis and treatment.

What remained unresolved. Specialty silos separated sexual well-being, mental health and social determinants.

Evidence anchors: [3,6]

2.0.4 Rights-based sexual/reproductive health, randomized evidence, STI programs and maternal-safety movements expanded the outcome beyond fertility.

Scientific consequence. Centered autonomy, prevention and population equity.

What remained unresolved. Legal and financing systems still restrict access and choice.

Evidence anchors: [3,6]

2.0.5 ART, genomics, fertility preservation, telehealth and cycle/wearable data expanded options and measurement.

Scientific consequence. Enabled individualized pathways and longitudinal data.

What remained unresolved. Commercial claims, intimate-data governance and unequal access create new risks.

Evidence anchors: [3,6]

2.0.6 Reproductive and sexual health is a life-course system linking biology, function, goals, relationships, rights, environment and healthcare.

Scientific consequence. Supports whole-person and cross-volume decisions.

What remained unresolved. Classification and research still undermeasure pleasure, coercion, disability and diverse identities.

Evidence anchors: ^[1,3]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Hypothalamic-pituitary-gonadal regulation

Core principle. Pulsatile neural-endocrine signaling coordinates gonadal steroid production, gametogenesis and reproductive cycles.

Control architecture. GnRH pulse patterns, LH/FSH feedback, gonadal hormones, inhibins, energy balance, stress and life stage.

Failure modes. Delayed/precocious puberty, anovulation, hypogonadism, infertility, sexual symptoms or bone/metabolic effects.

Observable consequences. History, pubertal/cycle pattern, timed hormones, imaging/genetics selectively and function.

Cross-system coupling. Brain, endocrine, metabolic, bone and psychosocial systems.

Evidence anchors: [3,9]

3.0.2 Oogenesis, follicular development and ovulation

Core principle. Finite follicles mature through coordinated endocrine and ovarian signals, with one or more oocytes potentially ovulated.

Control architecture. FSH/LH, granulosa/theca function, ovarian reserve, local growth factors, age and metabolic state.

Failure modes. Anovulation, diminished reserve, premature insufficiency, ovarian hyperstimulation or cystic states.

Observable consequences. Cycle tracking, ultrasound, timed progesterone, AMH/AFC as context-limited reserve proxies and fertility outcomes.

Cross-system coupling. Endocrine/metabolic, aging, genetics and maternal health.

Evidence anchors: [4,6,9]

3.0.3 Spermatogenesis and male reproductive function

Core principle. Germ cells mature in seminiferous tubules while accessory glands and ducts support transport and semen.

Control architecture. FSH/LH/testosterone, Sertoli/Leydig cells, temperature, genetics, toxins, infection and time.

Failure modes. Low count/motility/morphology, obstruction, endocrine failure, erectile/ejaculatory dysfunction or DNA damage.

Observable consequences. Repeated standardized semen analysis, hormones, exam/ultrasound/genetics when indicated and couple outcomes.

Cross-system coupling. Endocrine, environmental/occupational, infection and psychosocial systems.

Evidence anchors: [3,4,6]

3.0.4 Menstrual cycling and endometrial function

Core principle. Cyclic ovarian hormones coordinate endometrial growth, differentiation and shedding.

Control architecture. Follicular/ovulatory/luteal phases, prostaglandins, coagulation, uterine structure and systemic health.

Failure modes. Abnormal bleeding, amenorrhea, dysmenorrhea, implantation failure or anemia.

Observable consequences. Cycle/bleeding history, pregnancy test, blood count/hormones, imaging and endometrial sampling by risk.

Cross-system coupling. Hematologic, endocrine, uterine, pain and life-course systems.

Evidence anchors: [3,7,13]

3.0.5 Fertilization, implantation and early embryonic development

Core principle. Gametes unite, embryo develops and implants into receptive endometrium under precise temporal and molecular coordination.

Control architecture. Gamete competence, tubal transport, embryo genetics, endometrial state, immune tolerance and endocrine support.

Failure modes. Infertility, implantation failure, miscarriage, ectopic pregnancy or abnormal placenta-tion.

Observable consequences. Couple evaluation, ovulation/tubal/uterine/semen tests, genetics selectively and pregnancy localization.

Cross-system coupling. Maternal-fetal, immune, genetic and ethical domains.

Evidence anchors: [4,6,8]

3.0.6 Sexual response and genital function

Core principle. Desire, arousal, erection/lubrication, orgasm and satisfaction emerge from vascular, neurologic, hormonal, relational and contextual systems.

Control architecture. Autonomic/somatic pathways, blood flow, sensation, hormones, cognition, safety, relationship and medication effects.

Failure modes. Low desire, arousal/erectile, orgasmic, pain or penetration difficulties and distress.

Observable consequences. Respectful history, physical and targeted vascular/endocrine/neurologic assessment only when indicated.

Cross-system coupling. Cardiovascular, nervous, endocrine, mental health, relationship and pharmacology.

Evidence anchors: [2,3,17]

3.0.7 Pelvic floor and lower reproductive tract mechanics

Core principle. Muscle, fascia, nerves and organs support continence, sexual function, childbirth and pressure management.

Control architecture. Connective tissue, innervation, muscle coordination, pregnancy/childbirth, load and aging.

Failure modes. Prolapse, incontinence, pelvic pain, dyspareunia or defecatory dysfunction.

Observable consequences. History, pelvic exam, bladder/bowel/sexual function, strength/coordination and imaging/urodynamics selectively.

Cross-system coupling. Urinary, GI, musculoskeletal, maternal and rehabilitation systems.

Evidence anchors: [7,14]

3.0.8 Contraceptive physiology and reproductive autonomy

Core principle. Fertility can be modified through ovulation suppression, mucus/endometrial effects, gamete barriers, devices or sterilization.

Control architecture. Method mechanism, adherence, drug interactions, health conditions, preferences and access.

Failure modes. Unintended pregnancy, adverse effects, coercion, discontinuation or inequitable choice.

Observable consequences. Goals, medical eligibility, contraindications, interactions, use pattern and satisfaction.

Cross-system coupling. Ethics/rights, public health, pharmacology and life course.

Evidence anchors: [3,5,21]

3.0.9 Menopause and reproductive aging

Core principle. Declining follicular function changes cycles, vasomotor regulation, urogenital tissues, bone and cardiometabolic risk context.

Control architecture. Ovarian reserve depletion, endocrine adaptation, aging, health history and social meaning.

Failure modes. Symptoms, genitourinary syndrome, bone loss, sleep/mood effects or premature ovarian insufficiency.

Observable consequences. Clinical pattern, targeted labs when atypical, bone/cardiovascular risk and function.

Cross-system coupling. Endocrine, bone, cardiovascular, brain/behavior and aging.

Evidence anchors: [7,9,31]

3.0.10 Reproductive microbiome, infection and immune tolerance

Core principle. Mucosal barriers and microbial communities interact with sexual exposure, fertility and pregnancy while immunity must defend without disrupting reproduction.

Control architecture. Epithelium, pH, mucus, microbiota, innate/adaptive immunity and hormones.

Failure modes. STI, pelvic inflammatory disease, infertility, recurrent symptoms or adverse pregnancy outcomes.

Observable consequences. Syndrome-specific exam and validated pathogen tests; avoid interpreting microbiome profiles as diagnoses without utility.

Cross-system coupling. Infection, immune, maternal, oral/GI and public health.

Evidence anchors: [3,11]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Fecundability concept	$P(T = t) = (1 - p)^{t-1}p$	Under a constant per-cycle conception probability p , time to pregnancy follows a geometric model.	Real fecundability changes with age, timing, treatment and couple-specific factors.
Semen total motile count	$TMC = Volume \times Concentration \times Motile\ fraction$	Combines selected semen dimensions into a treatment-planning proxy.	It does not measure fertilizing capacity, DNA competence or couple-level fertility.
Diagnostic probability update	$Posttest\ odds = Pretest\ odds \times LR$	Reserve, ovulation or STI tests must be interpreted within pretest context.	Likelihood ratios and disease definitions must match the intended population.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within reproductive and sexual health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within reproductive and sexual health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[10]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[30]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[22]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[29]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the reproductive or sexual-health disorder, fertility state, functional phenotype, or rights-relevant care need, but cross-system processes and functional effects are retained.

5.0.1 Infertility and subfertility

Definition and phenotype. Failure to achieve pregnancy within a time frame appropriate to age and context, reflecting couple-level and individual factors.

Pathophysiology. Ovulatory, ovarian reserve, tubal/uterine, semen, sexual, genetic and unexplained mechanisms can coexist.

Determinants and modifiers. Age, infection, endometriosis, metabolic disease, toxins, cancer treatment, timing, access and social context.

Assessment. Concurrent couple evaluation: ovulation, anatomy/tubal patency, semen, history/exam and targeted endocrine/genetic tests.

Intervention logic. Cause-specific treatment, timed intercourse/insemination, surgery selectively and assisted reproduction with shared decisions.

Known limits and open questions. Ovarian reserve tests do not directly measure spontaneous fertility; “unexplained” reflects measurement limits.

Suite links. 03N maternal; 14 therapeutics; 15 ethics/equity.

Evidence anchors: ^[3,4,6]

5.0.2 Polycystic ovary syndrome

Definition and phenotype. Heterogeneous reproductive-metabolic syndrome with ovulatory dysfunction, hyperandrogenism and/or polycystic ovarian morphology after excluding mimics.

Pathophysiology. Neuroendocrine, ovarian and insulin-related pathways interact with genetics and environment.

Determinants and modifiers. Family liability, weight/metabolism, sleep, medications and social context.

Assessment. Life-stage-appropriate criteria, pregnancy exclusion, androgen and metabolic assessment; avoid ultrasound overdiagnosis in adolescents.

Intervention logic. Goals-based cycle/endometrial protection, fertility treatment, metabolic/weight support and symptom therapy.

Known limits and open questions. Phenotype boundaries and causal direction with obesity vary; label can create stigma.

Suite links. 03D endocrine/metabolic; 07 adolescence; 03N pregnancy.

Evidence anchors: [7,9,10]

5.0.3 Endometriosis and adenomyosis

Definition and phenotype. Endometrial-like tissue outside uterus or within myometrium associated with pain, inflammation and infertility in variable combinations.

Pathophysiology. Ectopic tissue, neuroangiogenesis, inflammation, fibrosis and central pain processes interact.

Determinants and modifiers. Genetics, menstruation/immune factors and diagnostic delay; causes remain incomplete.

Assessment. Clinical phenotype and ultrasound/MRI where useful; laparoscopy is not required for every case but may diagnose/treat selected disease.

Intervention logic. Analgesic/hormonal suppression, fertility planning, surgery selectively and multidisciplinary pain/rehabilitation.

Known limits and open questions. Lesion burden and pain correlate poorly; recurrence and optimal sequencing remain uncertain.

Suite links. 05 pain; 03E bowel; 03N fertility/pregnancy.

Evidence anchors: [6,7,16]

5.0.4 Uterine fibroids and structural bleeding

Definition and phenotype. Benign smooth-muscle tumors causing bleeding, pressure, pain or fertility/pregnancy effects depending on size/location.

Pathophysiology. Hormone-responsive clonal growth and extracellular matrix alter uterine mechanics and vascularity.

Determinants and modifiers. Age, ancestry, genetics, hormones, obesity and reproductive stage.

Assessment. Bleeding/anemia, pelvic exam, ultrasound and MRI selectively; fertility goals.

Intervention logic. Observation, medical bleeding control, procedures or surgery matched to symptoms, anatomy and reproductive goals.

Known limits and open questions. Symptom attribution and long-term comparative fertility outcomes vary.

Suite links. 03G anemia; 03N pregnancy; 14 procedures.

Evidence anchors: [6,7,13]

5.0.5 Abnormal uterine bleeding and amenorrhea

Definition and phenotype. Bleeding outside expected pattern or absence of menses from structural, ovulatory, endocrine, coagulation, pregnancy-related or iatrogenic causes.

Pathophysiology. Endometrial stability and hemostasis depend on ovulation, hormones, uterine structure and coagulation.

Determinants and modifiers. Age/life stage, pregnancy, endocrine/metabolic disease, bleeding disorders, drugs and stress/energy deficit.

Assessment. Urgency, pregnancy test, CBC, cycle/endocrine/coagulation context, imaging and sampling by age/risk.

Intervention logic. Stabilize severe bleeding; treat mechanism and protect endometrium; align fertility goals.

Known limits and open questions. Apps and self-report vary; normal ranges are life-stage and culture sensitive.

Suite links. 03G hematology; 03D endocrine; 03N pregnancy.

Evidence anchors: [3,7,13]

5.0.6 Sexually transmitted infections and reproductive-tract infection

Definition and phenotype. Pathogens transmitted primarily through sexual contact or ascending genital infection causing local/systemic disease and reproductive sequelae.

Pathophysiology. Mucosal invasion, inflammation, immune evasion and ascending spread can injure tubes, fetus or systemic organs.

Determinants and modifiers. Sexual networks, barrier use, vaccination, stigma, violence, screening access and AMR.

Assessment. Exposure/anatomic-site-specific NAAT/culture/serology and syndrome assessment; partner and pregnancy context.

Intervention logic. Vaccination, barrier/prevention services, effective antimicrobials, partner management and retesting according to pathogen.

Known limits and open questions. Syndromic treatment misses asymptomatic infection and microbiome tests can overmedicalize variation.

Suite links. 08 infection; 11 public health; 15 rights.

Evidence anchors: [3,11,12]

5.0.7 Pelvic inflammatory disease and tubal disease

Definition and phenotype. Upper-genital-tract inflammation/infection that may cause abscess, pain, ectopic pregnancy or infertility.

Pathophysiology. Ascending pathogens trigger inflammation, scarring and altered tubal transport.

Determinants and modifiers. Untreated STI, prior PID, access delays, procedures and microbiome/context.

Assessment. Low threshold clinical diagnosis when appropriate, pregnancy test, STI testing, ultrasound for abscess/alternative and severity.

Intervention logic. Prompt broad antimicrobial therapy, drainage/surgery for selected complications and prevention/partner care.

Known limits and open questions. No single test excludes mild disease; overtreatment and undertreatment both occur.

Suite links. 08 infection; 03N ectopic pregnancy; 14 AMR.

Evidence anchors: [3,7,11]

5.0.8 Erectile dysfunction

Definition and phenotype. Persistent difficulty attaining or maintaining erection adequate for desired sexual activity, with or without distress.

Pathophysiology. Vascular endothelial, neural, hormonal, medication, psychological and relational pathways can combine.

Determinants and modifiers. Cardiometabolic disease, smoking, pelvic surgery, drugs, stress, relationship and age.

Assessment. Sexual/medical history, focused exam, cardiometabolic risk and targeted endocrine/vascular tests when indicated.

Intervention logic. Risk-factor treatment, counseling/relationship support, PDE5 or device/injection/surgical options by safety and preference.

Known limits and open questions. It can be a vascular risk marker but is not specific; performance anxiety and organic disease coexist.

Suite links. 03A cardiovascular; 03D endocrine; 05/06 mental-relational.

Evidence anchors: [2,3,9]

5.0.9 Sexual pain and penetration disorders

Definition and phenotype. Pain, fear, muscle overactivity, tissue disease, neural sensitization or structural factors impair desired sexual activity.

Pathophysiology. Peripheral tissue/nerve input, pelvic floor guarding, central pain, trauma and expectation form feedback loops.

Determinants and modifiers. Infection/inflammation, menopause, childbirth, surgery, trauma, relationship safety and prior pain.

Assessment. Consent-centered history, external/pelvic exam only as appropriate, pain map, pelvic floor and psychosocial context.

Intervention logic. Treat tissue cause, pelvic rehabilitation, graded exposure/psychological/relationship support and selected medication/procedures.

Known limits and open questions. Broad labels can obscure specific causes; invasive exams can retraumatize.

Suite links. 05 pain; 06 relationships/trauma; 10 rehabilitation.

Evidence anchors: [3,14,16]

5.0.10 Pelvic organ prolapse and continence-related pelvic dysfunction

Definition and phenotype. Descent/support failure of pelvic organs with variable pressure, urinary, bowel and sexual symptoms.

Pathophysiology. Connective tissue, muscle, nerve and pressure-load changes follow childbirth, aging and other factors.

Determinants and modifiers. Parity/operative delivery, age, obesity, connective-tissue disease, chronic cough/constipation and work load.

Assessment. Symptom/function, standardized exam, bladder/bowel assessment and tests only if management requires.

Intervention logic. Pelvic floor training, pessary/assistive support or surgery based on goals and severity.

Known limits and open questions. Anatomic stage and symptom burden often disagree.

Suite links. 03F urinary; 03I connective tissue; 10 rehab.

Evidence anchors: [7,14,15]

5.0.11 Premature ovarian insufficiency and hypogonadism

Definition and phenotype. Loss or impairment of gonadal function earlier than expected, affecting fertility, symptoms, bone and systemic health.

Pathophysiology. Genetic, autoimmune, iatrogenic, infectious or idiopathic failure reduces hormones and gametogenesis.

Determinants and modifiers. Inherited variants, cancer therapy, surgery, immune disease and often unknown factors.

Assessment. Repeated timed hormones, pregnancy exclusion, etiologic testing/counseling and bone/cardiometabolic assessment.

Intervention logic. Hormone replacement when appropriate, fertility options, bone/vascular prevention and psychosocial support.

Known limits and open questions. Intermittent function and uncertain spontaneous fertility complicate counseling.

Suite links. 03D endocrine; 03I bone; 07 aging; 14 survivorship.

Evidence anchors: [6,7,9]

5.0.12 Menopause-related symptoms and genitourinary syndrome

Definition and phenotype. Vasomotor, sleep, mood, sexual and urogenital changes associated with menopausal transition and aging.

Pathophysiology. Thermoregulatory and tissue effects follow fluctuating/declining ovarian hormones with contextual modifiers.

Determinants and modifiers. Age/stage, symptom susceptibility, health history, stress, sleep and social meaning.

Assessment. Clinical pattern and differential; evaluate bleeding, bone/cardiovascular risk and contraindications.

Intervention logic. Behavioral/nonhormonal or systemic/local hormone therapy individualized to symptoms, risk and preference.

Known limits and open questions. Population-average risks should not replace individual absolute-risk discussion.

Suite links. 03D endocrine; 03A cardiovascular; 03I bone; 05 sleep/mood.

Evidence anchors: [7,9,30]

5.0.13 Reproductive cancers

Definition and phenotype. Cancers of ovary, endometrium, cervix, testis, prostate and other reproductive tissues with distinct biology.

Pathophysiology. Inherited/acquired genomic, hormonal, infectious and environmental pathways drive clonal disease.

Determinants and modifiers. Age, family variants, HPV, smoking, hormones and mostly unknown causes.

Assessment. Site-specific screening where recommended, symptoms, imaging/biopsy, staging and molecular testing by utility.

Intervention logic. Prevention/vaccination, surgery, radiation, systemic/targeted therapy and fertility/sexual rehabilitation.

Known limits and open questions. Screening net benefit differs by cancer; broad multi-cancer tests remain uncertain.

Suite links. 14 oncology; 08 HPV; 10 survivorship.

Evidence anchors: [12,19]

5.0.14 Gender-affirming reproductive and sexual health needs

Definition and phenotype. Reproductive organs, hormones, fertility goals and sexual health vary independently of gender identity and require individualized respectful care.

Pathophysiology. Hormonal/surgical interventions and baseline anatomy affect fertility, screening, sexual function and pregnancy possibilities.

Determinants and modifiers. Care access, discrimination, dysphoria, prior treatment, violence and social support.

Assessment. Anatomy- and goal-based history, organ-appropriate prevention, fertility counseling and mental-health support without pathologizing identity.

Intervention logic. Shared decision-making, evidence-based hormonal/surgical care, fertility preservation and inclusive preventive services.

Known limits and open questions. Long-term evidence is growing but incomplete; access and politicization distort care.

Suite links. 06 psychosocial; 15 rights/equity; 14 therapeutics.

Evidence anchors: ^[3,21,22]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current reproductive and sexual health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current reproductive and sexual health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Evidence anchors: ^[25]

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Reproductive and sexual history	Noninvasive	Consent-centered chronology of cycles, fertility attempts, pregnancies, sexual function, contraception, exposures, goals and safety.	Pretest model and person-valued outcomes.	Often highest-yield and low burden.	Stigma, language, recall, coercion and clinician assumptions.
Hormone assays	Minimally invasive sample	Immunoassay or mass-spectrometry measurement of reproductive and pituitary hormones.	Ovulatory/gonadal state, reserve proxies and endocrine causes.	Mechanistic support when timed and interpreted correctly.	Pulsatility, cycle timing, assay interference, binding proteins and reference-range misuse.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Semen analysis	Minimally invasive sample	Standardized macroscopic and microscopic assessment of volume, concentration, motility and morphology.	Male-factor probability and treatment planning.	Direct gamete-related information.	High within-person variability, abstinence/collection and laboratory variation; not fertility yes/no.
Pelvic and transvaginal ultrasound	Noninvasive/minimally invasive	Acronically imaging and Doppler of uterus, ovaries, follicles and pelvic structures.	Anatomy, follicle/ovulation, masses, bleeding and early pregnancy localization.	Dynamic, accessible and no ionizing radiation.	Operator, body habitus, cycle timing and incidental findings.
Tubal and uterine cavity testing	Minimally invasive	Contrast x-ray/ultrasound, saline infusion, hysteroscopy or laparoscopy.	Patency, cavity lesion and pelvic anatomy.	Actionable structural information.	Pain, infection, false occlusion, radiation or procedural risk.
Ovarian reserve testing	Minimally invasive/noninvasive	AMH, antral follicle count and cycle-specific FSH/estradiol.	Expected response to stimulation and ovarian follicle quantity proxy.	Useful for assisted-reproduction planning.	Poor direct prediction of spontaneous conception/oocyte quality and assay variation.
Ovulation and fertility tracking	Noninvasive/home	Urine LH/hormone metabolites, basal temperature, cervical mucus and wearable temperature/physiology algorithms.	Fertile-window estimate and cycle pattern.	Low-burden longitudinal self-observation.	Anovulation/irregular cycles, device algorithms, illness, sleep and false precision.
Genetic and carrier testing	Minimally invasive sample	Karyotype, targeted panels, sequencing and carrier/embryo testing.	Etiology, transmission risk and selected treatment decisions.	High value in defined syndromes and family planning.	Variants, incidental findings, ancestry bias, embryo-selection ethics and unequal access.
Sexual-function and pelvic-floor measures	Noninvasive/minimally invasive	Validated questionnaires, exam, pressure/EMG, sensory/vascular tests selectively.	Desire/arousal/erection and muscle coordination.	Clitoral/penis function symptoms and function.	Privacy, cultural norms, device intrusion and weak specificity.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Nocturnal penile tumescence and vascular testing	Noninvasive/minimally invasive	Clinically reference/rigidity sensing, Doppler or pharmacologic challenge.	Sleep erections and penile hemodynamic response.	Supports selected complex evaluation.	Sleep/device artifact, invasive challenge and cannot isolate psychological/organic causes perfectly.
STI and genital microbiology testing	Minimally invasive sample	NAAT, culture, microscopy, antigen and serology at relevant anatomic sites.	Pathogen and susceptibility evidence.	Highly actionable when appropriately sampled.	Window periods, colonization, site mismatch and resistance.
Endoscopy, laparoscopy and biopsy	Invasive	Direct visualization and tissue sampling of reproductive organs/pelvis.	Endometriosis, tumors, structural disease and pathology.	Direct diagnosis and treatment opportunity.	Anesthesia, bleeding, infection, adhesions and sampling.
Reproductive digital records and decision support	Software	Longitudinal cycles, medications, lab/imaging, embryos/gametes, procedures and outcomes.	Coordination, prediction and audit.	Can reduce fragmentation and preserve provenance.	Privacy, intimate-data misuse, algorithm bias and commercial sharing.

7.0.1 Reproductive and sexual history

Measured quantity and transducer. Consent-centered chronology of cycles, fertility attempts, pregnancies, sexual function, contraception, exposures, goals and safety.

Typical outputs. Pretest model and person-valued outcomes.

Strengths. Often highest-yield and low burden.

Limitations and artifacts. Stigma, language, recall, coercion and clinician assumptions.

Reference and validation strategy. Structured yet flexible tools, privacy and longitudinal confirmation.

Clinical/research maturity. Established core.

Evidence anchors: [3,4]

7.0.2 Hormone assays

Measured quantity and transducer. Immunoassay or mass-spectrometry measurement of reproductive and pituitary hormones.

Typical outputs. Ovulatory/gonadal state, reserve proxies and endocrine causes.

Strengths. Mechanistic support when timed and interpreted correctly.

Limitations and artifacts. Pulsatility, cycle timing, assay interference, binding proteins and reference-range misuse.

Reference and validation strategy. Traceability, timed repeats and clinical outcome validation.

Clinical/research maturity. Established.

Evidence anchors: [6,9]

7.0.3 Semen analysis

Measured quantity and transducer. Standardized macroscopic and microscopic assessment of volume, concentration, motility and morphology.

Typical outputs. Male-factor probability and treatment planning.

Strengths. Direct gamete-related information.

Limitations and artifacts. High within-person variability, abstinence/collection and laboratory variation; not fertility yes/no.

Reference and validation strategy. Repeated standardized collection and external quality assessment.

Clinical/research maturity. Established.

Evidence anchors: [4,6]

7.0.4 Pelvic and transvaginal ultrasound

Measured quantity and transducer. Acoustic imaging and Doppler of uterus, ovaries, follicles and pelvic structures.

Typical outputs. Anatomy, follicle/ovulation, masses, bleeding and early pregnancy localization.

Strengths. Dynamic, accessible and no ionizing radiation.

Limitations and artifacts. Operator, body habitus, cycle timing and incidental findings.

Reference and validation strategy. Standard protocols, inter-rater and surgical/pathology/outcome reference.

Clinical/research maturity. Established.

Evidence anchors: [6,7]

7.0.5 Tubal and uterine cavity testing

Measured quantity and transducer. Contrast x-ray/ultrasound, saline infusion, hysteroscopy or laparoscopy.

Typical outputs. Patency, cavity lesion and pelvic anatomy.

Strengths. Actionable structural information.

Limitations and artifacts. Pain, infection, false occlusion, radiation or procedural risk.

Reference and validation strategy. Protocol and operative/outcome correlation.

Clinical/research maturity. Established by indication.

Evidence anchors: ^[4,6]

7.0.6 Ovarian reserve testing

Measured quantity and transducer. AMH, antral follicle count and cycle-specific FSH/estradiol.

Typical outputs. Expected response to stimulation and ovarian follicle quantity proxy.

Strengths. Useful for assisted-reproduction planning.

Limitations and artifacts. Poor direct prediction of spontaneous conception/oocyte quality and assay variation.

Reference and validation strategy. Assay/platform and outcome-specific calibration.

Clinical/research maturity. Established with frequent overinterpretation.

Evidence anchors: ^[4,6]

7.0.7 Ovulation and fertility tracking

Measured quantity and transducer. Urinary LH/hormone metabolites, basal temperature, cervical mucus and wearable temperature/physiology algorithms.

Typical outputs. Fertile-window estimate and cycle pattern.

Strengths. Low-burden longitudinal self-observation.

Limitations and artifacts. Anovulation/irregular cycles, device algorithms, illness, sleep and false precision.

Reference and validation strategy. Reference ultrasound/hormones and prospective pregnancy/behavior utility.

Clinical/research maturity. Established narrow tests; wearables emerging.

Evidence anchors: [26,27]

7.0.8 Genetic and carrier testing

Measured quantity and transducer. Karyotype, targeted panels, sequencing and carrier/embryo testing.

Typical outputs. Etiology, transmission risk and selected treatment decisions.

Strengths. High value in defined syndromes and family planning.

Limitations and artifacts. Variants, incidental findings, ancestry bias, embryo-selection ethics and unequal access.

Reference and validation strategy. Analytic validity, counseling, phenotype and reproductive-outcome utility.

Clinical/research maturity. Established selected; expanding.

Evidence anchors: [6,28]

7.0.9 Sexual-function and pelvic-floor measures

Measured quantity and transducer. Validated questionnaires, exam, pressure/EMG, sensory/vascular tests selectively.

Typical outputs. Desire/arousal/erection/pain/function and muscle coordination.

Strengths. Centers symptoms and function.

Limitations and artifacts. Privacy, cultural norms, device intrusion and weak specificity.

Reference and validation strategy. Validated language/context and treatment responsiveness.

Clinical/research maturity. Established clinical tools; devices selected/research.

Evidence anchors: [3]

7.0.10 Nocturnal penile tumescence and vascular testing

Measured quantity and transducer. Circumference/rigidity sensing, Doppler or pharmacologic challenge.

Typical outputs. Sleep erections and penile hemodynamic response.

Strengths. Supports selected complex evaluation.

Limitations and artifacts. Sleep/device artifact, invasive challenge and cannot isolate psychological/organic causes perfectly.

Reference and validation strategy. Standard protocols and treatment/surgical outcome correlation.

Clinical/research maturity. Specialist.

Evidence anchors: [2,26]

7.0.11 STI and genital microbiology testing

Measured quantity and transducer. NAAT, culture, microscopy, antigen and serology at relevant anatomic sites.

Typical outputs. Pathogen and susceptibility evidence.

Strengths. Highly actionable when appropriately sampled.

Limitations and artifacts. Window periods, colonization, site mismatch and resistance.

Reference and validation strategy. Controls, site-specific performance and treatment/outcome follow-up.

Clinical/research maturity. Established.

Evidence anchors: [3,11]

7.0.12 Endoscopy, laparoscopy and biopsy

Measured quantity and transducer. Direct visualization and tissue sampling of reproductive organs/pelvis.

Typical outputs. Endometriosis, tumors, structural disease and pathology.

Strengths. Direct diagnosis and treatment opportunity.

Limitations and artifacts. Anesthesia, bleeding, infection, adhesions and sampling.

Reference and validation strategy. Surgical/pathology standards and patient-important outcomes.

Clinical/research maturity. Established by indication.

Evidence anchors: [7,19]

7.0.13 Reproductive digital records and decision support

Measured quantity and transducer. Longitudinal cycles, medications, lab/imaging, embryos/gametes, procedures and outcomes.

Typical outputs. Coordination, prediction and audit.

Strengths. Can reduce fragmentation and preserve provenance.

Limitations and artifacts. Privacy, intimate-data misuse, algorithm bias and commercial sharing.

Reference and validation strategy. Security, consent, interoperability and prospective utility.

Clinical/research maturity. Emerging/established infrastructure.

Evidence anchors: [23,24,25]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise reproductive and sexual health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise reproductive and sexual health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: ^[26]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

Evidence anchors: ^[25]

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [24,27]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [27]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [28]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Comprehensive sexuality education and preventive services	Prevention/p	Support informed consent, safer sex, vaccination, contraception and recognition of coercion/violence.	Improves knowledge and selected health outcomes without evidence that it increases sexual activity.	Political restriction, exclusion and unequal access.	Knowledge, service use, STI/unintended pregnancy, consent and equity.
Contraception	Prevention/a	Ulcer barrier, hormonal, intrauterine, emergency or permanent methods matched to health and preference.	Highly effective methods reduce unintended pregnancy; continuation depends on fit and autonomy.	Bleeding, metabolic/thrombotic or procedural risks vary; coercion is unacceptable.	Satisfaction, side effects, adherence/continuation, interactions and reproductive goals.
Fertility optimization and ovulation induction	Treatment	Correct endocrine/behavioral/exposure factors and stimulate ovulation when indicated.	Effective for defined ovulatory causes.	Multiple gestation, ovarian hyperstimulation, burden and inequitable access.	Ovulation, response, pregnancy/live birth and adverse outcomes.
Intrauterine insemination and assisted reproductive technology	Treatment	Facilitate gamete encounter or fertilization/embryo culture and transfer.	Improves pregnancy/live-birth probability in selected conditions; age and diagnosis dominate success.	Multiple pregnancy, procedures, emotional/financial burden, embryo/gamete ethics and disparities.	Cumulative live birth, maternal/neonatal outcomes, burden and long-term child follow-up.
Fertility preservation	Preventive/a	Cryopreserve gametes, embryos or tissue before gonadotoxic treatment or anticipated decline.	Established for selected patients; probability depends on age, method and future use.	Delay, procedure risk, cost, storage and uncertain future success.	Mature gamete/embryo yield, later use, live birth and consent disposition.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Hormonal and nonhormonal reproductive therapy	Treatment	Modify cycles, bleeding, pain, androgen/menopause symptoms or replace deficient hormones.	Benefits are indication-specific and preference-sensitive.	Thrombosis, metabolic, mood, bone, cancer-context and drug-interaction risks vary.	Symptoms, bleeding, bone/metabolic/vascular risk and adverse effects.
Pelvic-floor rehabilitation and sexual therapy	Rehabilitation/whole person	Improve muscle coordination, pain, continence and sexual function with physical and psychosexual approaches.	Useful for selected pelvic pain, prolapse, postpartum and sexual dysfunction.	Trauma sensitivity, access and variable protocols.	Pain/function, continence, sexual goals, participation and distress.
Surgery and minimally invasive procedures	Structural treatment	Treat fibroids, endometriosis, obstruction, prolapse, tumors, sterilization or reconstructive needs.	Effective for defined anatomy and goals.	Bleeding, infection, adhesions, fertility loss, recurrence and recovery.	Symptoms, fertility/function, pathology, complications and recurrence.
STI vaccination, treatment and partner services	Prevention/treatment	Prevent/system vaccine-sensitive infection and interrupt transmission with testing, therapy and partner care.	Strong for defined pathogens and public-health pathways.	Stigma, AMR, privacy and loss to follow-up.	Cure/reinfection, resistance, partner treatment and equity.
Cancer prevention, treatment and sexual/fertility survivorship	Oncology/rehabilitation	Screen/vaccinate where evidence supports and integrate fertility/sexual function into cancer care.	Improves survival and quality when timely and stage-appropriate.	Treatment-induced infertility, menopause, sexual dysfunction and access.	Cancer outcomes, fertility, sexual function, symptoms and participation.
Rights-based and trauma-informed care	Governance/clinical	Protect privacy, consent, bodily autonomy, noncoercion, inclusive language and safety.	Essential to quality and uptake; rights violations directly harm health.	Institutional coercion, discrimination and data misuse.	Patient-reported respect/safety, informed choice, complaint/incident and equity metrics.

9.0.1 Comprehensive sexuality education and preventive services

Target and mechanism. Support informed consent, safer sex, vaccination, contraception and recognition of coercion/violence.

Evidence and expected outcome. Improves knowledge and selected health outcomes without evidence that it increases sexual activity.

Risks, burden, and equity. Political restriction, exclusion and unequal access.

Response monitoring. Knowledge, service use, STI/unintended pregnancy, consent and equity.

Evidence anchors: [3,12]

9.0.2 Contraception

Target and mechanism. Use barrier, hormonal, intrauterine, emergency or permanent methods matched to health and preference.

Evidence and expected outcome. Highly effective methods reduce unintended pregnancy; continuation depends on fit and autonomy.

Risks, burden, and equity. Bleeding, metabolic/thrombotic or procedural risks vary; coercion is unacceptable.

Response monitoring. Satisfaction, side effects, adherence/continuation, interactions and reproductive goals.

Evidence anchors: [3,5]

9.0.3 Fertility optimization and ovulation induction

Target and mechanism. Correct endocrine/behavioral/exposure factors and stimulate ovulation when indicated.

Evidence and expected outcome. Effective for defined ovulatory causes.

Risks, burden, and equity. Multiple gestation, ovarian hyperstimulation, burden and inequitable access.

Response monitoring. Ovulation, response, pregnancy/live birth and adverse outcomes.

Evidence anchors: [4,6]

9.0.4 Intrauterine insemination and assisted reproductive technology

Target and mechanism. Facilitate gamete encounter or fertilization/embryo culture and transfer.

Evidence and expected outcome. Improves pregnancy/live-birth probability in selected conditions; age and diagnosis dominate success.

Risks, burden, and equity. Multiple pregnancy, procedures, emotional/financial burden, embryo/gamete ethics and disparities.

Response monitoring. Cumulative live birth, maternal/neonatal outcomes, burden and long-term child follow-up.

Evidence anchors: [4,6,8]

9.0.5 Fertility preservation

Target and mechanism. Cryopreserve gametes, embryos or tissue before gonadotoxic treatment or anticipated decline.

Evidence and expected outcome. Established for selected patients; probability depends on age, method and future use.

Risks, burden, and equity. Delay, procedure risk, cost, storage and uncertain future success.

Response monitoring. Mature gamete/embryo yield, later use, live birth and consent disposition.

Evidence anchors: [6,19]

9.0.6 Hormonal and nonhormonal reproductive therapy

Target and mechanism. Modify cycles, bleeding, pain, androgen/menopause symptoms or replace deficient hormones.

Evidence and expected outcome. Benefits are indication-specific and preference-sensitive.

Risks, burden, and equity. Thrombosis, metabolic, mood, bone, cancer-context and drug-interaction risks vary.

Response monitoring. Symptoms, bleeding, bone/metabolic/vascular risk and adverse effects.

Evidence anchors: [2,7,9]

9.0.7 Pelvic-floor rehabilitation and sexual therapy

Target and mechanism. Improve muscle coordination, pain, continence and sexual function with physical and psychosexual approaches.

Evidence and expected outcome. Useful for selected pelvic pain, prolapse, postpartum and sexual dysfunction.

Risks, burden, and equity. Trauma sensitivity, access and variable protocols.

Response monitoring. Pain/function, continence, sexual goals, participation and distress.

Evidence anchors: [3,14]

9.0.8 Surgery and minimally invasive procedures

Target and mechanism. Treat fibroids, endometriosis, obstruction, prolapse, tumors, sterilization or reconstructive needs.

Evidence and expected outcome. Effective for defined anatomy and goals.

Risks, burden, and equity. Bleeding, infection, adhesions, fertility loss, recurrence and recovery.

Response monitoring. Symptoms, fertility/function, pathology, complications and recurrence.

Evidence anchors: [6,7,29]

9.0.9 STI vaccination, treatment and partner services

Target and mechanism. Prevent vaccine-sensitive infection and interrupt transmission with testing, therapy and partner care.

Evidence and expected outcome. Strong for defined pathogens and public-health pathways.

Risks, burden, and equity. Stigma, AMR, privacy and loss to follow-up.

Response monitoring. Cure/reinfection, resistance, partner treatment and equity.

Evidence anchors: [3,11,12]

9.0.10 Cancer prevention, treatment and sexual/fertility survivorship

Target and mechanism. Screen/vaccinate where evidence supports and integrate fertility/sexual function into cancer care.

Evidence and expected outcome. Improves survival and quality when timely and stage-appropriate.

Risks, burden, and equity. Treatment-induced infertility, menopause, sexual dysfunction and access.

Response monitoring. Cancer outcomes, fertility, sexual function, symptoms and participation.

Evidence anchors: [14,19,20]

9.0.11 Rights-based and trauma-informed care

Target and mechanism. Protect privacy, consent, bodily autonomy, noncoercion, inclusive language and safety.

Evidence and expected outcome. Essential to quality and uptake; rights violations directly harm health.

Risks, burden, and equity. Institutional coercion, discrimination and data misuse.

Response monitoring. Patient-reported respect/safety, informed choice, complaint/incident and equity metrics.

Evidence anchors: [3,21,28]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can ovarian aging be predicted precisely for an individual?	Age and reserve markers relate to follicle quantity and treatment response.	Timing of natural fertility decline or menopause with narrow individual certainty.	Markers do not directly measure oocyte competence and trajectories vary.	Diverse longitudinal cohorts with repeated assays and reproductive outcomes.
Do reproductive microbiome tests improve outcomes?	Microbial communities vary with anatomy, hormones and behavior.	Stable causal endotypes and effective targeted interventions beyond pathogen testing.	Sampling, compositional methods and confounding create unstable associations.	Locked prospective intervention trials with fertility/symptom outcomes.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within reproductive and sexual health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can ovarian aging be predicted precisely for an individual?

Established knowledge. Age and reserve markers relate to follicle quantity and treatment response.

Unresolved issue. Timing of natural fertility decline or menopause with narrow individual certainty.

Why current evidence is insufficient. Markers do not directly measure oocyte competence and trajectories vary.

Research that would change confidence. Diverse longitudinal cohorts with repeated assays and reproductive outcomes.

Evidence anchors: ^[4,6]

10.0.2 Do reproductive microbiome tests improve outcomes?

Established knowledge. Microbial communities vary with anatomy, hormones and behavior.

Unresolved issue. Stable causal endotypes and effective targeted interventions beyond pathogen testing.

Why current evidence is insufficient. Sampling, compositional methods and confounding create unstable associations.

Research that would change confidence. Locked prospective intervention trials with fertility/symptom outcomes.

Evidence anchors: ^[27]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within reproductive and sexual health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[27]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[19]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[24,26]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Couple-level fertility model	Can transparent multimodal models improve time-to-pregnancy counseling without deterministic labeling?	Prospective diverse couples with repeated semen/cycle/exposure data and cumulative live-birth outcomes.	Calibrated decision benefit and honest uncertainty across age and access groups.	High discrimination driven by age but no improved decisions.
0. Privacy-preserving reproductive sensor standard	Can cycle and sexual-health devices share auditable raw/quality data without exposing intimate information?	Open schema, threat modeling, hardware validation and user-governed consent study.	Interoperable, minimal-data system with validated intended use and low privacy harm.	Commercial telemetry remains opaque or re-identifiable.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for reproductive and sexual health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Couple-level fertility model

Question. Can transparent multimodal models improve time-to-pregnancy counseling without deterministic labeling?

Design. Prospective diverse couples with repeated semen/cycle/exposure data and cumulative live-birth outcomes.

Success criterion. Calibrated decision benefit and honest uncertainty across age and access groups.

Failure or stop criterion. High discrimination driven by age but no improved decisions.

Evidence anchors: [4,27]

11.0.2 0. Privacy-preserving reproductive sensor standard

Question. Can cycle and sexual-health devices share auditable raw/quality data without exposing intimate information?

Design. Open schema, threat modeling, hardware validation and user-governed consent study.

Success criterion. Interoperable, minimal-data system with validated intended use and low privacy harm.

Failure or stop criterion. Commercial telemetry remains opaque or reidentifiable.

Evidence anchors: [23,25,26]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for reproductive and sexual health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: ^[22]

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[27]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[22]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03N	Maternal, Fetal and Perinatal Health	Conception, implantation, pregnancy outcomes and fertility-treatment effects continue into maternal-fetal care.	pregnancy is possible, planned or established.
03D	Endocrine and Metabolic Health	HPG, thyroid, prolactin, insulin and obesity affect cycles, fertility and sexual function.	hormonal or metabolic phenotype is central.
03A	Cardiovascular Health	Vascular health affects erectile function, pregnancy risk and hormonal/contraceptive decisions.	sexual symptoms or treatment changes vascular risk.
06	Psychosocial and Relational Health	Consent, relationships, identity, trauma, stigma and cultural context are integral.	distress or goals cannot be explained biologically alone.
15	Ethics, Law, Rights, Equity and Governance	Autonomy, fertility technology, genetic testing, abortion/contraception law and intimate-data rights.	care or research involves reproductive choice or contested governance.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Fecundability	Probability of conception within a menstrual cycle under specified exposure and measurement conditions.
Ovarian reserve	Quantity-related proxy of remaining follicular pool and expected stimulation response; not a direct measure of natural fertility or oocyte quality.
Sexual health	State of physical, emotional, mental and social well-being in relation to sexuality, requiring autonomy, safety and rights.
Reproductive autonomy	Capacity to make and act on informed reproductive decisions free from coercion or discrimination.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.